Python Basics and Programming Concepts

- Introduction to Python
  - Python is a versatile, high-level, open-source language loved by programmers globally.
  - Suitable for beginners with no prior coding experience; can be used in AI, machine learning, data science, and web development (Django).
  - Python syntax closely resembles English, making it easy to understand and learn.
  - Programs consist of code (instructions) written to communicate with machines, which only understand binary (0s and 1s).
  - Translators (interpreters/compilers) convert Python code into machine code.
  - Python supports multi-platform compatibility and has frameworks like Django and pip.
- Writing Basic Programs
  - Writing and running first Python program printing text to console.
  - Usage of print() function to show output.
  - Multiple print statements for displaying variables and outputs.
  - Understanding syntax requirements:Strings are enclosed in quotes (single, double, or triple).Variables store values and can be manipulated.Use variable names without quotes to print their stored values.
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- Operators and Expressions
  - Arithmetic operators perform mathematical operations (+, -, *, /, %, **).
  - Relational (comparison) operators compare two values, returning boolean (True/False) results.
  - Assignment operators assign values and support compound assignments (+=, -=, *=, /=, %=, **=).
  - Logical operators (not, and, or) manipulate boolean values to combine conditions.
  - Python is case sensitive, so keywords like True, False, None must start with uppercase letters.
- Comments and Code Readability
  - Comments use # for single line or triple quotes for multi-line comments.
  - Comments are ignored during execution, used to explain code.
  - Use meaningful variable names and neat code structure to improve readability.
  - Use shortcuts in editors like VS Code to comment/uncomment multiple lines.
- Practice Examples Overview
  - Taking two numbers as input and printing their sum.
  - Calculating area of a square from input side length.
  - Computing average of two float numbers.
  - Comparing two numbers and printing True if one is greater or equal.
  - Demonstrated can combine inputs, type conversions, arithmetic, relational and logical operators.
- Setting Up Environment
  - Installation steps for Python on Windows and macOS:Download Python installer (python.org), check 'Add Python to PATH' box during installation.Verify installation by running python3 --version in terminal.
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  - Visual Studio Code (VS Code) is recommended for coding:Download based on OS and system architecture (64-bit or 32-bit).Save Python files with .py extension.Use built-in Run button to execute Python scripts.
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- Variables and Data Types
  - Variables: Named memory locations storing data; values can change over time.
  - Python automatically infers data types. Common primary data types:int (integer values, whole numbers including negatives and zero)float (decimal values)str (string/text, enclosed in quotes)bool (boolean type with True/False)NoneType (None indicates absence of value)
    - int (integer values, whole numbers including negatives and zero)
    - float (decimal values)
    - str (string/text, enclosed in quotes)
    - bool (boolean type with True/False)
    - NoneType (None indicates absence of value)
  - Variable naming rules:Case-sensitive, no special characters except underscore.Meaningful, simple, short names recommended.
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  - Use type() function to check data types of variables.
- Type Conversion
  - Python automatically converts compatible types (implicit conversion):e.g., int to float during arithmetic involving both.
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  - Explicit type casting available to convert data types manually (int(), float(), str()).
  - Input values read via input() function are always strings by default.
  - Convert input to desired type explicitly using type casting before calculation.
- Input and Output Interaction
  - input() pauses program, waits for user input, stores entered value in a variable.
  - Use prompts in input() to tell users what to enter.
  - Convert string input to necessary data types (int, float) before calculations.
  - Combine input, variable storage, and print statements to create interactive programs.

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